

2 (a) During respiration, respiratory substrates such as glucose are used as a source of energy to produce ATP molecules.

(i) Cells can maintain a different concentration of ions within the cell compared with the extracellular environment.

It is estimated that animals may use 50% of ATP made from respiration to maintain higher concentrations of some ions inside cells.

Name the mechanism that can maintain a higher concentration of ions inside a cell.

..... [1]

- (ii) Only a proportion of energy released by the respiration of glucose is used to make ATP.
- The energy yield from the respiration of glucose in aerobic conditions is 2870 kJ mol^{-1} .
 - When ATP is hydrolysed to ADP, it releases 30.5 kJ mol^{-1} of energy.
 - It is estimated that 31 moles of ATP are made per mole of glucose respired.

Calculate the percentage efficiency of glucose respiration.

Show your working.

..... % [2]

(iii) The energy that is **not** converted to ATP during respiration is released as heat energy.

State **one** homeostatic use of this heat energy in mammals.

..... [1]

(iv) State the term used to describe the reaction that results in the production of ATP during the Krebs cycle.

..... [1]

(b) When a person exercises, power is generated.

Power is measured in joules per second (J s^{-1}).

The power generated when a person exercises will vary depending on the type and intensity of exercise. More power is generated when the intensity of exercise increases.

An experiment was carried out to determine whether increasing the intensity of exercise in a healthy human male caused a change in the value of the respiratory quotient (RQ).

- The man had a balanced diet.
- The RQ was calculated at rest.
- He carried out exercise for the same length of time on four separate days.
- The intensity of exercise was increased over the four days, generating different powers.
- The RQ was calculated for the four different powers generated.

The results are shown in Fig. 2.1.

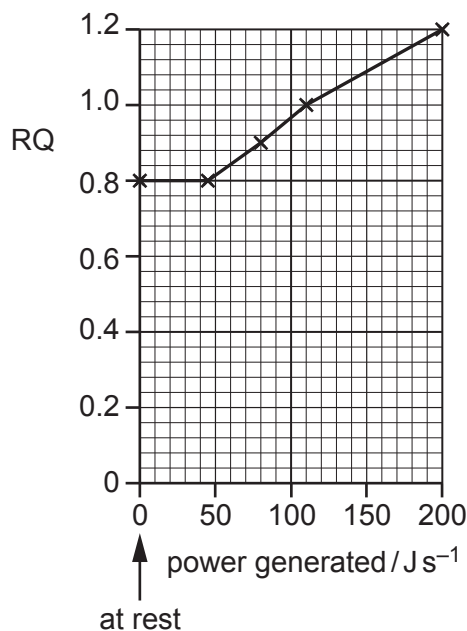


Fig. 2.1

With reference to Fig. 2.1, suggest what can be deduced from:

the RQ at rest

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the RQ when the power generated is 110 J s^{-1}

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the RQ when the power generated is 200 J s^{-1} .

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[4]

- (c) Explain why a person continues to breathe deeply and at a higher rate for some time after the person has stopped exercising.

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[3]

[Total: 12]